FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS – GRADE 10 | SUB-STRAND 3.1: STATISTICS 1	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS
This is your Final Explanation document. Use it as a model to help you write a complete, high-quality answer to the driving question. Read each section carefully, study the exemplar responses, and then craft your own answer using evidence from your investigations, data collection, and class activities. Write in full sentences. Use correct mathematical vocabulary. Always connect your answer back to Mama Wanjiku's canteen problem — and to the bigger picture of how statistics helps Kenyan decision-makers solve real problems.

Section 1: Why Raw Data Alone Cannot Solve Problems	
Explain why Mama Wanjiku's memory-based guesses failed. What is raw data, and why is it not yet useful on its own? What must happen to raw data before it can help anyone make a decision?	Mama Wanjiku served over 120 learners every lunch break, but she relied on her memory to decide how much food to prepare each day. Human memory is not designed to track, count, and compare large amounts of information accurately — it tends to remember the most recent or most dramatic events rather than true patterns. As a result, she often ran out of ugali and sukuma wiki (the most popular meal) and was left with too much githeri (the least popular meal), wasting money and leaving learners hungry. Raw data is a collection of unorganised facts or numbers — for example, a long list of 120 meal choices written down in the order learners gave them. On its own, this list looks like 'meaningless noise.' You cannot see patterns, totals, or comparisons just by staring at an unorganised list. Before raw data becomes useful, it must be organised, summarised, and displayed. The Grade 10 learners did exactly this: they transformed Mama Wanjiku's chaotic lunch orders into a frequency distribution table and graphs, which immediately revealed clear patterns she had never seen before.

Section 2: Organising Data — Frequency Distribution Tables and Tally Charts	
Describe how you organise raw data into a frequency distribution table. What are tallies and frequencies? How does a completed table help	A frequency distribution table organises raw data into categories and shows how often each category occurs. To build one, you first list all the possible categories (in Mama Wanjiku's case: ugali with sukuma wiki, chapati, githeri, rice and beans, and mandazi). Then, for every data entry in your raw list, you place a tally mark (/) next to the correct category. Every fifth tally crosses through the previous four (////), making it easy to count groups of five. Finally, you count the tally marks for each category and record the total as the frequency.

you see patterns in the data? Use the canteen data as your example.	For the canteen data from 120 learners over one week, the completed table showed: – Ugali with sukuma wiki — frequency: ~56 per day (the highest) – Chapati — frequency: moderate – Githeri — frequency: the lowest – Rice and beans — frequency: moderate – Mandazi — frequency: low Once the table is complete, the pattern jumps out immediately: ugali with sukuma wiki has by far the highest frequency. This single table gave Mama Wanjiku more reliable information than years of guessing from memory. A frequency distribution table turns a chaotic list into an organised summary that is easy to read and compare.

Section 3: Visualising Data — Bar Charts and Pie Charts	
Explain how to draw a bar chart and a pie chart from a frequency table. What does each type of graph show? How did the graphs help Mama Wanjiku understand her canteen data more easily than the table alone?	Graphs translate numbers into pictures, making patterns even easier to see. A BAR CHART uses rectangular bars of equal width, where the height of each bar represents the frequency of that category. To draw one correctly: (1) draw and label the horizontal axis with your categories (the five meals), (2) draw and label the vertical axis with a suitable scale for frequency, (3) draw bars for each category — ensuring the bars do not touch each other for categorical data, and (4) give the graph a clear title. When the Grade 10 learners drew the bar chart for Mama Wanjiku's canteen data, the bar for ugali with sukuma wiki towered above all the others, instantly showing it was the most popular meal. Githeri had the shortest bar, confirming it was least popular. A PIE CHART shows each category as a slice of a circle, where the size of each slice represents that category's share of the total. To calculate the angle for each slice, use: Angle = (Frequency ÷ Total) × 360°. For ugali with sukuma wiki: (56 ÷ 120) × 360° ≈ 168°, which is nearly half the pie — or about 47% of all orders. This confirmed what the bar chart showed, but in a different visual way: ugali dominates the menu by proportion. Using both graphs together gave Mama Wanjiku a complete picture. The bar chart made comparison between meals easy; the pie chart showed the share of the whole. Together, they made the evidence undeniable.

Section 4: Measures of Central Tendency — Mean, Median, and Mode	
Define mean, median, and mode. Show how each is calculated using the canteen data. Explain what each measure tells Mama Wanjiku and why using all three together gives a more complete picture than using just one.	Measures of central tendency are single numbers that summarise a whole data set by describing its 'centre' or 'typical value.' There are three main measures: MEAN (Average): Add up all the values and divide by the number of values. For the canteen data, the mean number of ugali portions sold per day = (total portions sold across the week) ÷ 5 days = 290 ÷ 5 = 58 portions per day. This tells Mama Wanjiku that, on average, she should prepare at least 58 portions of ugali each day.

MEDIAN (Middle value): Arrange all values in order from smallest to largest, then find the middle value. If there is an even number of values, find the average of the two middle values. For the canteen ugali data (e.g., 54, 56, 57, 60, 63), the median is 57. The median is useful because it is not affected by extreme values (outliers) the way the mean can be.

MODE (Most frequent value): The value that appears most often in the data set. For the canteen data, the mode is 60 portions — this was the most commonly recorded daily total for ugali. The mode is especially useful for non-numerical categories; for example, 'ugali with sukuma wiki' is the modal meal choice.

Using all three measures together gives Mama Wanjiku a fuller picture: the mean (58) helps her plan average daily stock; the median (57) confirms the typical day is not distorted by unusual days; and the mode (60) tells her the most common single-day total to prepare for. Relying on just one measure could lead to under- or over-stocking on any given day.

Section 5: Answering the Driving Question — Statistics as a Decision-Making Tool for Kenya

Now bring everything together. How do frequency tables, graphs, and measures of central tendency transform raw data into a decision-making tool? Answer the driving question fully. Then extend your answer: beyond the school canteen, how could the same statistical tools help other Kenyan decision-makers — such as a health worker in Kisumu, a matatu operator in Nairobi, or a farmer in the Rift Valley — solve real problems?

Mama Wanjiku's canteen problem is a perfect example of a universal challenge: having lots of information but no way to understand it. Raw data — an unorganised list of 120 meal choices — was invisible information. It existed, but it could not be used. By applying the statistical tools learned in this sub-strand, the Grade 10 learners transformed that invisible information into clear, actionable knowledge.

Here is how the tools worked together as a system:

1. A frequency distribution table organised the chaos into categories and counts.
2. A bar chart and pie chart made the patterns visible at a glance.
3. The mean, median, and mode gave precise, reliable numbers for daily planning.

The result? Mama Wanjiku now knows to prepare approximately 58 portions of ugali with sukuma wiki every day (the mean), that 60 portions is the most common daily demand (the mode), and that the typical day sits at 57 portions (the median). She also knows that githeri is the least popular meal and should be prepared in far smaller quantities. She can reduce waste, save money, and ensure every hungry learner is served.

The same tools can solve problems far beyond one school canteen:

- A health worker in Kisumu collecting data on malaria cases per week can use a frequency table and line graph to spot which weeks have the highest infection rates and prepare medicine stocks accordingly.
- A matatu operator in Nairobi can record passenger numbers at different times of day, calculate the mean and mode, and decide when to add extra vehicles on the busiest routes.
- A maize farmer in the Rift Valley can record rainfall data over several seasons, find the median and mean rainfall, and use a bar chart to decide the best planting time.

In every case, the process is the same: collect raw data → organise it into a frequency table → display it in a graph → calculate measures of central tendency → make an informed decision. Statistics is not just a mathematics topic — it is a thinking tool that helps Kenyan communities solve real problems with evidence instead of guesswork.

	So, to answer the driving question directly: We can turn raw data from our school and community into useful statistics by organising it in frequency tables, displaying it in bar charts and pie charts, and calculating the mean, median, and mode. When we do this, hidden patterns become visible, and decision-makers like Mama Wanjiku can act with confidence, fairness, and accuracy — not just memory and guesswork.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Understanding of Raw Data and the Need for Organisation	Clearly explains why raw, unorganised data is not useful on its own. Accurately describes what must happen to data before it can inform decisions. Uses the canteen phenomenon as a specific, well-developed example. Demonstrates deep understanding of why Mama Wanjiku's memory failed.	Explains that raw data needs to be organised and gives a correct reason. References the canteen context. Minor gaps in depth or precision.	Shows a basic or partial understanding that data needs to be organised. Explanation is vague or does not clearly connect to the canteen phenomenon. May confuse raw data with organised data.
Construction and Interpretation of Frequency Distribution Tables and Graphs	Accurately describes how to construct a frequency distribution table using tallies and frequencies. Correctly explains how to draw both a bar chart and a pie chart, including axis labels, scale, titles, and angle calculations. Interprets both graphs correctly using canteen data values.	Describes the construction of a frequency table and at least one graph type correctly. Calculations and labels are mostly accurate. Interpretation is reasonable but may lack full detail.	Attempts to describe a frequency table or graph but contains errors in construction, labelling, or calculation. Interpretation is limited or incorrectly linked to the data.
Calculation and Interpretation of Mean, Median, and Mode	Correctly defines and calculates mean, median, and mode using canteen data. Clearly explains what each measure tells Mama Wanjiku about her daily planning. Explains why using all three together gives a more complete picture than using just one measure.	Correctly defines and calculates at least two of the three measures. Provides a reasonable interpretation of each in context. May not fully explain the advantage of using all three.	Attempts to define or calculate at least one measure of central tendency. Definitions or calculations contain errors. Interpretation is absent or weakly connected to the canteen context.
Connecting Statistics to Real Kenyan Decision-Making Contexts	Provides a thorough, well-reasoned answer to the driving question. Extends the application of statistical tools to at least two other Kenyan contexts (e.g., health, transport, farming). Explanations are specific, realistic, and demonstrate transfer of learning beyond the classroom.	Answers the driving question adequately and extends to at least one other Kenyan context. The extension example is relevant but may lack specific detail or a clear explanation of how the statistical tools would be applied.	Partially answers the driving question. Extension to other contexts is absent, vague, or not clearly connected to statistical tools. Response stays narrowly within the canteen example.
Use of Mathematical Language and Logical Structure	Consistently uses correct and precise statistical vocabulary throughout (e.g., frequency, tally, mean, median, mode, bar chart, pie chart, class interval,	Uses correct statistical vocabulary in most places. Response is generally organised and readable. Some mathematical working is shown. Minor	Uses everyday language instead of mathematical vocabulary, or uses terms incorrectly. Response is poorly organised or difficult to follow. Mathematical

	distribution). Response is logically organised, flows clearly from one idea to the next, and uses full sentences. Mathematical working is clearly shown where required.	errors in terminology or structure do not obscure meaning.	working is missing or incomplete.